

RESEARCH INTERESTS

My research focuses on improving the reliability, safety, and performance of LLM agents through studying their inner workings. My recent work aims at prying the LLM black box open so we can “fix it like a car.” In particular, my research makes models:

- **Interpretable.** I develop mechanistic interpretability methods that reveal the function-specific structures inside LLMs.
- **Controllable.** I intervene in these structures to steer LLM behavior: disable distraction, control reasoning, edit knowledge.
- **Modular.** I show that the mechanisms can become reusable components, towards a future of composable, modular AI.

Keywords: Large Language Models (LLMs); LLM Post-Training; Mechanistic Interpretability; LLM Reliability, Safety & Control; LLM Reasoning & In-Context Learning; Agentic Systems.

EDUCATION

University of Toronto, Ph.D. in Computer Science. 2020.05 — 2025.01

Advisor: Gerald Penn.

Committee: Gerald Penn, Frank Rudzicz, David Duvenaud, Ewan Dunbar, and Cynthia Rudin.

Thesis: Towards the Renormalization of Transformer Language Models.

University of Toronto, Master of Science (M.Sc.) in Computer Science. 2018.09 — 2020.04

Advisor: Gerald Penn.

University of Toronto, Honours Bachelor of Science (B.Sc.) in Computer Science. 2014.09 — 2018.05

ACADEMIC POSITIONS

Ubiquitous Knowledge Processing (UKP) Lab, TU Darmstadt. 2025.01 — Present

Postdoctoral Researcher.

Darmstadt, Germany

Advisor: Iryna Gurevych.

PROFESSIONAL EXPERIENCE

Microsoft Research. Research Scientist Intern. 2024.06 — 2024.09

Mentors: Xingdi Yuan, Tong Wang, Hamidreza Saghir, and Amir H. Abdi.

Montreal, Canada

Public Health Agency of Canada. FSWEF Student Research Scientist. 2020.02 — 2024.04

Mentors: Victoria Ng, Erin E. Rees, and Simon de Montigny.

Toronto, Canada

Vector Institute. 2021.09 — 2024.12

Faculty Affiliate Researcher.

Toronto, Canada

iNAGO Corporation. 2017.11 — 2018.09

Research Intern.

Toronto, Canada

IBM Canada. 2017.01 — 2017.10

Software Engineering Intern.

Markham, Canada

HONOURS AND AWARDS

- Journal-to-Conference (J2C) Certification, TMLR. 2025
- Outstanding Paper Award, ACL 2025. 2025
- Outstanding Reviewer Award, ARR May 2025/EMNLP 2025. 2025
- Spotlight, ICLR 2024. 2024
- Federal Student Work Experience Program (FSWEF). 2020 — 2024

Last updated: May 2026.

- Vector Research Faculty Affiliate Award. 2023
- Ph.D. Conference Travel Award. 2022
- Vector Postgraduate Affiliate Award. 2021, 2022
- St. Michael’s College Silver Medal. 2018
- Dean’s List — University of Toronto. 2014 — 2017

TEACHING EXPERIENCE

Instructor at TU Darmstadt.

- Introduction to Large Language Models. Winter 2025

Instructor at the University of Toronto.

- CSC485/2501 Introduction to Computational Linguistics. Fall 2024

Head Teaching Assistant at the University of Toronto.

- CSC401/2511 Natural Language Computing. Winter 2024
- CSC485/2501 Introduction to Computational Linguistics. Fall 2023

Teaching Assistant at the University of Toronto.

- CSC485/2501 Introduction to Computational Linguistics. Fall 2019, 2020, 2021, 2022
- CSC401/2511 Natural Language Computing. Winter 2021, 2022, 2023
- CSC263: Data Structures and Analysis. Winter 2017, 2018, 2019, 2020
- CSC236: Introduction to the Theory of Computation. Fall 2015, Summer 2021
- CSC148: Introduction to Computer Science. Fall 2018, Summer 2019

PUBLICATIONS

Refereed Conference Publications

*: Equal contribution. †: Corresponding author. —: Supervised students.

1. All Circuits Lead to Rome: Rethinking Functional Anisotropy in Circuit and Sheaf Discovery for LLMs.
Xi Chen*, Mingyu Jin*, **Jingcheng Niu***, Yutong Yin, Jinman Zhao, Bangwei Guo, Dimitris N. Metaxas, Zhaoran Wang, Yutao Yue, Gerald Penn.
To appear in *Forty-Third International Conference on Machine Learning (ICML)*, 2026.
2. Dynamic Granularity in the Wild: Differentiable Sheaf Discovery with Joint Computation Graph Pruning.
Lei Yu*, **Jingcheng Niu***, Zining Zhu, Xi Chen, Gerald Penn.
In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
3. Tiny Budgets, Big Gains: Parameter Placement Strategy in Parameter Super-Efficient Fine-Tuning.
Jinman Zhao, Xueyan Zhang, Jiaru Li, **Jingcheng Niu**, Yulan Hu, Erxue Min, Gerald Penn.
In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
4. Llama See, Llama Do: A Mechanistic Perspective on Contextual Entrainment and Distraction in LLMs.
Jingcheng Niu, Xingdi Yuan, Tong Wang, Hamidreza Saghir, Amir H. Abdi.
In *Proceedings of the Annual Conference of the Association for Computational Linguistics (ACL)*, 2025.
🏆 **Outstanding Paper Award**. Top 0.3% of submissions.
5. Attenuating Catastrophic Forgetting of Token-Level Knowledge in Contrastive Sentence Embeddings with Attentive Pooling.
Saifei Liao, **Jingcheng Niu**, Gerald Penn.
In *Proceedings of the 38th Canadian Conference on Artificial Intelligence (CAIAC)*, 2025.
6. ConTempo: Temporally Contrastive Learning for Temporal Relation Extraction.
Jingcheng Niu, Saifei Liao, Victoria Ng, Simon de Montigny, Gerald Penn.
In *Findings of the Association for Computational Linguistics (ACL Findings)*, 2024.

7. What does the Knowledge Neuron Thesis Have to do with Knowledge?
Jingcheng Niu, Andrew Liu, Zining Zhu, Gerald Penn.
 In *International Conference on Learning Representations (ICLR)*, 2024.
 **Spotlight**. Top 5% of submissions.
8. Chinese Quantifier Scope, Concord, and Lexical Resource Semantics.
Jingcheng Niu, Xinyu Kang, Pascal Hohmann, Gerald Penn.
 In *Proceedings of the International Conference on Head-Driven Phrase Structure Grammar (HPSG)*, 2022.
9. Does BERT Rediscover a Classical NLP Pipeline?
Jingcheng Niu, Wenjie Lu, Gerald Penn.
 In *Proceedings of the International Conference on Computational Linguistics (COLING)*, 2022. Oral Presentation.
10. Temporal Histories of Epidemic Events (THEE): A Case Study in Temporal Annotation for Public Health.
Jingcheng Niu, Erin E. Rees, Victoria Ng, Gerald Penn.
 In *Proceedings of the Language Resources and Evaluation Conference (LREC)*, 2020.
11. Rationally Reappraising ATIS-based Dialogue Systems.
Jingcheng Niu, Gerald Penn.
 In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.

Journal Publications

12. Hypothesis-Driven Feature Manifold Analysis in LLMs via Supervised Multi-Dimensional Scaling.
Federico Tiblias, Irina Bigoulaeva, **Jingcheng Niu**, Simone Balloccu, Iryna Gurevych.
 Transactions on Machine Learning Research (TMLR), 2026.
13. Illusion or Algorithm? Investigating Memorization, Emergence, and Symbolic Processing in In-Context Learning.
Jingcheng Niu, Subhabrata Dutta, Ahmed Elshabrawy, Harish Tayyar Madabushi, Iryna Gurevych.
 Transactions on Machine Learning Research (TMLR), 2025.
 **Journal-to-Conference (J2C) Certification**. Top 5% of submissions.
14. A Call for an Ethical Framework When Using Social Media Data for Artificial Intelligence Applications in Public Health Research.
 Jean-Philippe Gilbert, Victoria Ng, **Jingcheng Niu**, Erin E. Rees.
 Canada Communicable Disease Report (CCDR), 2020.
15. Application of Natural Language Processing Algorithms for Extracting Information from News Articles in Event-Based Surveillance.
 Victoria Ng, Erin E. Rees, **Jingcheng Niu**, Abdelhamid Zaghouel, Homeira Ghiasbeglou, Adrian Verster.
 Canada Communicable Disease Report (CCDR), 2020.

Refereed Workshop Publications

16. Discourse Information for Document-Level Temporal Dependency Parsing.
Jingcheng Niu, Victoria Ng, Erin E. Rees, Simon de Montigny, Gerald Penn.
 In *Proceedings of the Workshop on Computational Approaches to Discourse (CODI)*, 2023.
17. Using Roark-Hollingshead Distance to Probe BERT’s Syntactic Competence.
Jingcheng Niu, Wenjie Lu, Eric Corlett, Gerald Penn.
 In *Proceedings of the BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP (BlackboxNLP)*, 2022.
18. Statistically Evaluating Social Media Sentiment Trends towards COVID-19 Non-Pharmaceutical Interventions with Event Studies.
Jingcheng Niu, Erin E. Rees, Victoria Ng, Gerald Penn.
 In *Proceedings of the Sixth Social Media Mining for Health Workshop and Shared Task (SMM4H)*, 2021.
19. Machine Learning Identification of Self-reported COVID-19 Symptoms from Tweets in Canada.
 Jean-Philippe Gilbert, **Jingcheng Niu**, Simon de Montigny, Victoria Ng, Erin E. Rees.
 In *Proceedings of the International Workshop on Health Intelligence (W3PHAI)*, 2021.
20. Grammaticality and Language Modelling.
Jingcheng Niu, Gerald Penn.
 In *Proceedings of the Workshop on Evaluation and Comparison of NLP Systems (Eval4NLP)*, 2020.

Refereed Industry Track Conference Publications

21. Bringing the State-of-the-Art to Customers: A Neural Agent Assistant Framework for Customer Service Support.
Stephen Obadinma, Faiza Khan Khattak, Shirley Wang, Tania Sidhom, Elaine Lau, Sean Robertson, **Jingcheng Niu**, Winnie Au, Alif Munim, Karthik Raja K. Bhaskar, Bencheng Wei, Iris Ren, Waqar Muhammad, Erin Li, Bukola Ishola, Michael Wang, Griffin Tanner, Yu-Jia Shiah, Sean X. Zhang, Kwesi P. Apponsah, Kanishk Patel, Jaswinder Narain, Deval Pandya, Xiaodan Zhu, Frank Rudzicz, Elham Dolatabadi.
In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: Industry Track (EMNLP Industry Track)*, 2022.

Preprints & Work Under Review

22. Harder the Task, Sparser the Representation: Sparsity as a Learning Signature of Capability in LLMs.
Mingyu Jin, Yutong Yin, **Jingcheng Niu**[†], Qingcheng Zeng, Wujiang Xu, Mengnan Du, Wei Cheng, Zhaoran Wang, Tianlong Chen, Dimitris N. Metaxas.
Under Review, 2026.
23. Many Circuits, One Mechanism: Input Variation and Evaluation Granularity in Circuit Discovery.
Alireza Makou, **Jingcheng Niu**, Subhabrata Dutta, Iryna Gurevych.
Under Review, 2026.
24. Skill-RAG: Failure-State-Aware Retrieval Augmentation via Hidden-State Probing and Skill Routing.
Kai Wei, Raymond Li, Xi Zhu, Zhaoqian Xue, Jiaojiao Han, **Jingcheng Niu**, Fan Yang.
Under Review, 2026.
25. Lagrangian Dual Ascent for Improving Sparse Dictionary Learning.
Raymond Li, **Jingcheng Niu**, Chuyuan Li, Fengyu Cai, Gabriel Murray, Giuseppe Carenini.
Under Review, 2026.
26. A Mechanistic View of Spatial Reasoning in Vision-Language Models.
Jie Cai, **Jingcheng Niu**, Zhejian Zhou, Fan Yang, Jesse Thomason.
Under Review, 2026.
27. Agent MechSuits: Mechanistic Subspace Safety Steering for Multi-Turn CLI Agents.
Weidi Luo, Qiming Zhang, Mingyu Jin, Jie Cai, Yihao Quan, Chaowei Xiao, **Jingcheng Niu**, Zhen Xiang.
Under Review, 2026.

INVITED TALKS

1. Prying the LLM Blackbox Open Towards Interpretable, Controllable & Composable AI.
Invited Talk, State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences. April 2026.
2. Prying the LLM Blackbox Open Towards Interpretable, Controllable & Composable AI.
Invited Talk, Purdue University. March 2026.
3. Prying the LLM Blackbox Open Towards Interpretable, Controllable & Composable AI.
Invited Talk, Saint Louis University. March 2026.
4. Prying the LLM Blackbox Open Towards Interpretable, Controllable & Composable AI.
Cardiff NLP seminar, Cardiff University. March 2026.
5. We Owe Language Models an Explanation.
The Past, Present, and Future of the UKP Lab, Workshop for the Royal Society Milner Award. November 2025.
6. How Do LLMs Reason about Time?
The Workshop on AI-based Methods for the Humanities, BIFOLD Institute. September 2025.
7. Llama See, Llama Do: A Mechanistic Perspective on Contextual Entrainment and Distraction in LLMs.
ACL 2025 Outstanding Paper Talk, AI TIME. August 2025.
8. Dissecting Language Models: From Black Boxes to Interpretable AI.
UKP-CIS Joint Invited Talk Series, TU Darmstadt and LMU Munich. July 2024.
9. Dissecting Language Models: From Black Boxes to Interpretable AI.
Linguistics Colloquium, Goethe-Universität Frankfurt. May 2024.

10. Dissecting Language Models to Understand Their Internal Workings.
AICamp Toronto. March 2024.
11. Using Roark-Hollingshead Distance to Probe BERT’s Syntactic Competence.
Vector Research Symposium. February 2024.
12. Syntax and Language Modelling.
ACL Student Research Seminar, University of Toronto. July 2023.
13. Does BERT Rediscover a Classical NLP Pipeline?
Vector Research Symposium. March 2023.
14. Chinese Quantifier Scope, Concord, and Lexical Resource Semantics.
Linguistics Colloquium, Goethe-Universität Frankfurt. October 2022.
15. Understanding Public Sentiment about COVID-19 Non-Pharmaceutical Interventions through Event Studies.
Vector Research Symposium. February 2021.

ACADEMIC SERVICE

Area Chair

- 2026: EACL, ACL, EMNLP.
- 2025: EMNLP.

Reviewer

- 2026: ICLR, NeurIPS.
- 2025: ACL, EMNLP (**Outstanding Reviewer**), NAACL, ICLR, NeurIPS, AAAI, CODI Workshop.
- 2024: ACL, EACL, EMNLP, NAACL, CODI Workshop.
- 2023: IJCNLP, EACL, ACL, EMNLP, FEVER Workshop.
- 2022: ACL, EMNLP.

MENTORSHIPS AND SUPERVISION

Academic Mentorship

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| <ul style="list-style-type: none"> • Federico Tiblias, Ph.D. student at the UKP Lab, TU Darmstadt.
Day-to-day supervision; Ph.D. advisor: Iryna Gurevych.
<i>Topic</i>: Geometric and causal structure in latent representations. | 2025 — Present |
| <ul style="list-style-type: none"> • Alireza Makou, Ph.D. student at the UKP Lab, TU Darmstadt.
Day-to-day co-supervision with Subhabrata Dutta; Ph.D. advisor: Iryna Gurevych.
<i>Topic</i>: Circuit discovery. | 2025 — Present |
| <ul style="list-style-type: none"> • Mingyu Jin, Ph.D. student at Rutgers University, New Brunswick.
External mentorship and research collaboration.
<i>Topic</i>: Mechanistic interpretability. | 2025 — Present |
| <ul style="list-style-type: none"> • Xi Chen, Undergraduate student at the University of Toronto.
Informal mentorship and research collaboration.
<i>Topic</i>: Circuit discovery. | 2024 — 2025 |
| <ul style="list-style-type: none"> • Saifei Liao, Undergraduate student at the University of Toronto.
Informal mentorship and research collaboration.
<i>Topic</i>: Contrastive learning. | 2022 — 2023 |

Thesis Students

- Hanifi Akdag, M.Sc. student at TU Darmstadt. 2025
Master's Thesis at the UKP Lab, TU Darmstadt.
Joint with Alireza Makou.
- Xiaoyan Xue, M.Sc. student at TU Darmstadt. 2025
Master's Thesis at the UKP Lab, TU Darmstadt.
Joint with Hovhannes Tamoyan.
- Tam Truong, M.Sc. student at TU Darmstadt. 2025
Master's Thesis at the UKP Lab, TU Darmstadt.
Joint with Hovhannes Tamoyan.
- Jonas Schulz, Undergraduate student at TU Darmstadt. 2025
Undergraduate Thesis at the UKP Lab, TU Darmstadt.
Joint with Hovhannes Tamoyan.

KEY SKILLS

Large Language Models	Pre-training & in-context learning dynamics; post-training (SFT, RLHF, DPO, PEFT: LoRA, adapters); agentic systems (tool use, multi-turn agents, RAG); multi-modal (vision-language models).
LLM Reliability & Control	Hallucination & distraction mitigation, representation steering, model editing, agent safety.
Mechanistic Interpretability	Circuit discovery, activation patching & causal tracing, attention head ablation, feature manifold analysis, sparse autoencoders (SAEs), probing, knowledge neurons.
Programming	Python, PyTorch, JAX, TensorFlow, HuggingFace Transformers, distributed training (FSDP, DeepSpeed) on Slurm clusters.
Machine Learning & Mathematics	Deep learning, reinforcement learning, Bayesian modeling, statistical learning theory, information theory, optimization.